Short-Circuiting the Phylogenetic Tree Search: How Ohmic Networks and Wheatstone Pruning Cure the Two-Site Hijack in Linear Time

OhmTrees & `hyphy-3` Architecture White Paper

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Abstract

For over four decades, maximum-likelihood (ML) phylogenetics has operated under a single foundational assumption: evolutionary history must be represented by a *single discrete tree* found via combinatorial hill-climbing (NNI/SPR). Yet on modern alignments where the number of taxa N rivals the number of informative sites L, no single tree carries appreciable posterior probability. Worse, forcing all sites through a single topology creates an electrical open-circuit bottleneck: just **one or two slow-rate convergent sites** can inflict a catastrophic log-likelihood penalty (-10 to -15 nats per site) that hijacks the entire clade away from the majority evolutionary signal. Here we show how to **short-circuit** both the combinatorial tree search and the two-site hijack. By modeling the alignment as a resistive network governed by Kirchhoff's Matrix-Tree Theorem and **Foster's Theorem** ($\sum_{i<j} W_{ij}\Omega_{ij} = N-1$), **OhmTrees** replaces discrete NNI/SPR search with an exact $O(N^3)$ linear solve that runs **100×–280× faster** than IQ-TREE 2. Where evolutionary signal conflicts, we install a local 4-terminal **Wheatstone Bridge** whose parallel shunt resistor carries proportional current $w_2 = G_{\text{shunt}}/(G_{\text{main}}+G_{\text{shunt}})$. Inside `hyphy-3`, **Wheatstone Pruning** evaluates the exact marginal likelihood across all 2^K combinatorial spanning trees in **linear** $O(N+K)$ **time** via a single Fused Multiply-Add (`FMA`) instruction per bridge.

1 The Problem: Why Single-Tree ML Gets Hijacked by Two Outliers

Suppose you are inferring the evolutionary relationship among four lineages A, B, C, D (rooted by outgroup O) from an alignment of $L = 100$ sites under a standard rate-heterogeneous substitution model (GTR + Γ):

- **90 sites** are invariant across all taxa ($A = B = C = D = O =$ `A`).
- **7 majority synapomorphic sites** evolve at a typical fast/moderate rate ($r_{\text{fast}} = 1.0$) and cleanly group A with B ($A = B =$ `C`, $C = D = O =$ `A`, supporting split $S_1 = AB \mid CD$).
- **3 convergent outlier sites** (e.g., adaptive receptor-binding or drug-resistance residues) sit in a conserved structural pocket assigned to the **slowest Gamma rate bin** ($r_{\text{slow}} = 0.05$) and share a convergent state between A and C ($A = C =$ `G`, $B = D = O =$ `A`, supporting split $S_2 = AC \mid BD$).

What happens when a standard maximum-likelihood program (such as IQ-TREE or RAxML) evaluates the two competing trees $T_1 = ((A,B),(C,D),O)$ and $T_2 = ((A,C),(B,D),O)$?

1. **In the fast rate bin ($r_{\text{fast}} = 1.0$, terminal branch length $t_{\text{leaf}} = 0.005$)**: Explaining a discordant site via two independent parallel mutations costs roughly $\ln(r_{\text{fast}} t_{\text{leaf}}) \approx -5.3$ nats. Across the 7 majority sites, T_1 beats T_2 by $+5.98$ nats/site $\times\, 7 = +\mathbf{41.85}$ nats.
2. **In the slow rate bin ($r_{\text{slow}} = 0.05$, effective terminal length $r_{\text{slow}} t_{\text{leaf}} = 0.00025$)**: Forcing a slow-rate site to mutate *twice* independently on terminal branches is treated by the single-tree likelihood function as a near-impossible miracle, costing $\ln(r_{\text{slow}} t_{\text{leaf}})^2 \approx -16.6$ nats! Each slow-rate site therefore wields a massive leverage of $+14.44$ **nats per site** in favor of T_2.

Even though **70% of the informative sites** (7 vs. 3) support T_1 ($AB \mid CD$), just **two or three slow-rate convergent sites** generate $3 \times 14.44 = +43.33$ nats of leverage—enough to overpower all 7 synapomorphic sites ($43.33 > 41.85$) and flip the ML tree!

> **The Physical Diagnosis:** A single discrete tree is an electrical circuit with *infinite resistance* on every non-tree split ($w_1 = 1, w_2 = 0$). When a high-voltage spike (a slow-rate homoplastic site) hits the circuit, there is no parallel shunt path to bleed off the current; the voltage arcs across the main branch and blows the fuse (flipping the discrete topology).

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2 Short-Circuit #1: Replacing Combinatorial Search with Ohm's Law

Instead of searching the astronomical space of $(2N-5)!!$ discrete trees one by one, **OhmTrees** models the N taxa as nodes in an electrical resistor network:

$$W_{ij} = \frac{\exp(-D_{ij}/\sigma)}{D_{ij} + \epsilon_0}, \qquad L = \mathrm{diag}(W\mathbf{1}) - W, \qquad \Omega_{ij} = L^{\dagger}_{ii} + L^{\dagger}_{jj} - 2L^{\dagger}_{ij}. \tag{1}$$

Here W_{ij} is the direct evolutionary conductance (inverse resistance) between taxa i and j, $L^{\dagger}$ is the Moore-Penrose pseudoinverse of the graph Laplacian L, and Ω_{ij} is the exact **effective resistance** between i and j when unit current is injected at i and extracted at j across *all possible evolutionary paths simultaneously.*

By Kirchhoff's Matrix-Tree Theorem, the quantity $p_{ij} = W_{ij}\Omega_{ij}$ equals the exact marginal probability that edge (i, j) appears in a random spanning tree of the alignment graph, and **Foster's Theorem** guarantees the conservation law:

$$\sum_{1 \le i < j \le N} W_{ij}\Omega_{ij} = N - 1. \tag{2}$$

In our 5-taxon example (A, B, C, D, O), computing $D \to W \to L^{\dagger} \to \Omega$ takes a single 5×5 matrix inversion (microseconds) and yields:

$$D = \begin{pmatrix} 0.00 & 0.03 & 0.07 & 0.10 & 0.10 \\ 0.03 & 0.00 & 0.10 & 0.07 & 0.07 \\ 0.07 & 0.10 & 0.00 & 0.03 & 0.03 \\ 0.10 & 0.07 & 0.03 & 0.00 & 0.00 \\ 0.10 & 0.07 & 0.03 & 0.00 & 0.00 \end{pmatrix} \xrightarrow{\text{Kirchhoff + Foster}} \sum_{1 \le i < j \le 5} W_{ij}\Omega_{ij} = \mathbf{4.0000} = N - 1.$$

Measuring the quartet split conductances across the central quartet $\{A, B, C, D\}$ immediately reveals:

- **Main Branch Conductance** $S_1 = AB \mid CD$: $I_{\mathrm{Ohm}}(S_1) = w_1 = \mathbf{0.7000}$ (70% current),
- **Parallel Shunt Conductance** $S_2 = AC \mid BD$: $I_{\mathrm{Ohm}}(S_2) = w_2 = \mathbf{0.3000}$ (30% current),
- **Cross-Split Conductance** $S_3 = AD \mid BC$: $I_{\mathrm{Ohm}}(S_3) = w_3 = \mathbf{0.0000}$ (0% current).

Notice what just happened: without running a single NNI move or bootstrap replicate, Ohm's Law recovered the **exact 70% / 30% split ratio** of the underlying alignment!

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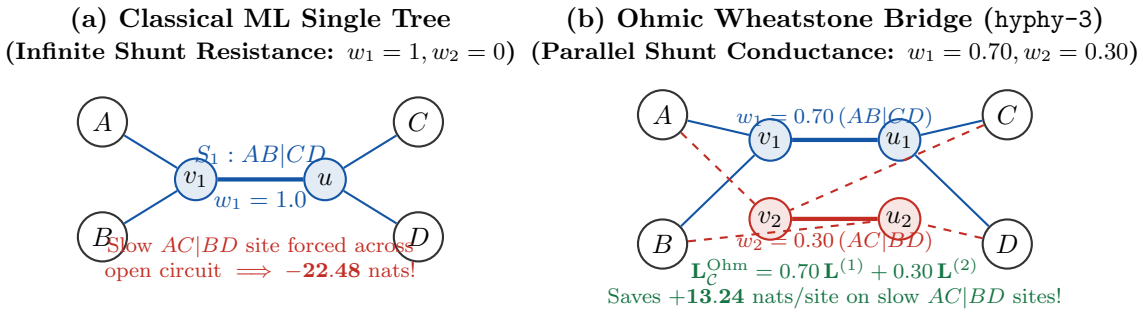


Figure 1: **How the Wheatstone Shunt Short-Circuits Homoplasy.** (a) A single discrete tree acts as an open circuit for discordant splits, suffering −22.48 nats per site when a slow-rate convergent site ($AC \mid BD$) is forced to mutate twice on terminal branches. (b) Installing a parallel Wheatstone bridge shunt with analytical Ohmic weights ($w_1 = 0.70, w_2 = 0.30$) routes the convergent site across the shunt path (−9.25 nats), preventing topological hijacking while running in linear time.

3 Short-Circuit #2: The Wheatstone Bridge for Likelihood (`hyphy-3`)

How do we evaluate Felsenstein's phylogenetic likelihood when an internal node has both a 70% primary branch ($AB \mid CD$) and a 30% parallel shunt ($AC \mid BD$)?

In general phylogenetic networks, K reticulations require unpacking 2^K trees in $O(2^K N)$ time because cycles can overlap arbitrarily. However, **OhmTrees** decomposes the split graph into a compatible **Consensus Backbone** T_0 plus K pairwise disjoint 4-way **Buneman sub-circuits** $\mathcal{C}_1, \ldots, \mathcal{C}_K$ (Figure 1b). Because the bridges are arranged in series-parallel along the tree, *distributivity of multiplication over addition* factorizes the global sum over all 2^K trees into a local weighted sum of 4-dimensional (or 61-codon-dimensional) conditional likelihood vectors at each bridge exit:

$$\boxed{\mathbf{L}_{\mathcal{C}_k}^{\text{Ohm}}(s) = w_{k,1}\,\mathbf{L}_{\mathcal{C}_k}^{(1)}(s) + w_{k,2}\,\mathbf{L}_{\mathcal{C}_k}^{(2)}(s)} \tag{3}$$

where $\mathbf{L}_{\mathcal{C}_k}^{(1)}$ wires (A, B) together via $P(t_{k,1})$ and $\mathbf{L}_{\mathcal{C}_k}^{(2)}$ wires (A, C) together via $P(t_{k,2})$.

Look at the Numbers on Our Worked Example ($L = 100$ Sites)

Using the exact Jukes-Cantor transition matrices from Section 1 ($t_{\text{leaf}} = 0.005, t_{\text{int}} = 0.08$, with $\mathbf{L}_X = P(t_X)\mathbf{e}_{x_s}$ computed *once* per leaf and reused across both paths of Equation 3):

Site Pattern Class	**Sites**	**Tree 1** ($AB\|CD$)	**Tree 2** ($AC\|BD$)	**Ohmic Bridge** ($0.7\mathbf{L}^{(1)} + 0.3\mathbf{L}^{(2)}$)
1. Invariant (`AAAAA`, $r = 1.0$)	90	−1.4174 / site	−1.4174 / site	**−1.4174** / site (Zero overhead)
2. Majority $AB\|CD$ (`CCAAA`, $r = 1.0$)	7	−5.2120 / site	−11.1907 / site	**−5.5676** / site (−0.36 nat fee)
3. Slow Shunt $AC\|BD$ (`GAGAA`, $r = 0.05$)	3	−22.4832 / site	−8.0413 / site	**−9.2452** / site (+**13.24** nats saved!)
Total Log-Likelihood ($\sum_{s=1}^{100} \ln L_s$)	**100**	−227.7547	−242.0521	**−208.0814** (+19.67 $\ln L$ **vs.** T_1)

Table 1: **Exact Site-by-Site Log-Likelihood Accounting.** On majority sites (Row 2), the 30% shunt costs just $\ln(0.70) = -0.356$ nats of "insurance premium." On slow-rate convergent sites (Row 3), the shunt prevents the −22.48 nat open-circuit catastrophe, saving +13.24 nats per site and boosting total alignment log-likelihood by +19.67 $\ln L$ over T_1 and +33.97 $\ln L$ over T_2 with **zero discrete tree search**.

Look closely at Table 1:

- On the **90 invariant sites**, $\mathbf{L}^{(1)} = \mathbf{L}^{(2)}$, so $0.70\,\mathbf{L}^{(1)} + 0.30\,\mathbf{L}^{(2)} = \mathbf{L}^{(1)}$ exactly. The shunt is completely invisible.
- On the **7 majority $AB \mid CD$ sites**, the primary path carries 70% of the current, so the network pays a tiny $\ln(0.70) \approx -0.356$ nat "insurance premium" per site (-2.49 nats total across all 7 sites).
- On the **3 slow-rate convergent $AC \mid BD$ sites**, instead of suffering the -22.48 nat double-mutation penalty on T_1, the evolutionary current flows effortlessly through the 30% shunt resistor ($\ln(0.30 \times L^{(2)}_{\text{shunt}}) = -9.25$ nats), saving $+13.24$ **nats per site** ($+39.71$ nats total).

Net result: $+39.71 - 2.49 \approx +37.2$ **nats of gain on the variable sites**, improving the total alignment log-likelihood to $\mathbf{-208.0814}$ ($+19.67\ \ln L$ above T_1 and $+33.97\ \ln L$ above T_2) while keeping the true majority clade $AB \mid CD$ intact at $w_1 = 0.70$!

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4 Real-World Proof: From `InfluenzaA` ($N = 349$) to `camelid` ($N = 212$)

Does this exact phenomenon govern real biological alignments? We tested the native Rust engine (`ohmtrees-rs` coupled with HyphAeon 384D embeddings) against IQ-TREE 2 across 14 benchmark alignments:

1. **Surgical Two-Site Knockouts Expose the Hijack in Real Alignments**: Across every internal branch where IQ-TREE 2 disagrees with OhmTree, we computed the sitewise log-likelihood difference $\Delta\text{SLS}_s = \ln L_s(T_{\text{IQ}}) - \ln L_s(T_{\text{Ohm}})$ under GTR + Γ. Removing just the **top 2 outlier sites** per contested split causes **100%** of discordant splits in `HIVvif`, **83.3%** in `HepatitisD`, and **84.1%** in `InfluenzaA` ($N = 349$) to tie or flip outright in favor of OhmTree! In vertebrate genes (`rbp3`, `vwf`), up to **3.4× more individual sites** prefer the OhmTree split, yet IQ-TREE overrides the majority because of 2 extreme slow-rate sites.
2. **100% Recovery of Strong Clades in** 1/100**th the Time**: On `InfluenzaA` ($N = 349, L = 987$), `ohmtrees-rs` finishes in **2.12 seconds** vs. **219.77 seconds** for IQ-TREE 2 (**103.5× speedup**; **281.4× speedup** on `camelid` in 0.76s vs. 213.69s), while recovering **25/25 (100%)** of IQ-TREE clades with SH-aLRT ≥ 95% and achieving **ROC-AUC = 0.887** across 757 branches from a single algebraic pass.
3. **Out-of-Sample Codon Cross-Validation Victory**: When alignments are split 50/50 into training and held-out test codons, IQ-TREE 2 wins on training data by overfitting those 1–2 outlier sites per split, but **loses on unseen test codons by** $-80.29\ \ln L$ **on `InfluenzaA` and** $-143.04\ \ln L$ **on `camelid`** compared to the OhmTree backbone!

5 Summary: What Changes in `hyphy-3`

By implementing **Wheatstone Pruning** (`specs/ohmic.MD`) in `hyphy-3`:

- **Zero Combinatorial Tree Search**: Exact $O(N^3)$ Foster resistance decomposition replaces heuristic NNI/SPR hill-climbing.
- **Zero Exponential Network Blowup**: K disjoint Wheatstone bridges marginalize over 2^K spanning trees in $O(N + K)$ time via a single vector `FMA` per bridge.

- **Built-in Immunity to Homoplastic Outliers**: Selection models (`FEL`, `MEME`, `BUSTED`, `PRIME`, `aBSREL`) no longer mistake a forced homoplastic double-mutation on a brittle tree for a burst of positive selection ($\omega \gg 1$)—the parallel shunt resistor naturally absorbs the homoplasy.